

Lateral-Zigzag PN Junction Enabled High-Efficiency Silicon Micro-Ring Modulator Working at 100Gb/s

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Abstract—We propose and demonstrate a novel lateral-zigzag PN junction for silicon modulator working in C and O bands. The designed PN junction exhibits characteristics of large bandwidth, high modulation efficiency, and easy fabrication. For demonstration, it is embedded in a 220 nm silicon rib waveguide on a 2 μm thick buried oxide layer to form a micro-ring modulator. The measured 3 dB electro-optic bandwidth of the fabricated modulator is 42 GHz while the resistor-capacitor (RC) bandwidth is measured and calculated to be 174 GHz. Clear and open eye diagrams are obtained for 50 Gb/s non-return zero (NRZ) and 100 Gb/s four-level pulse amplitude modulation (PAM4) signals. The modulation power consumption is only 14 fJ/bit.

Index Terms—High-efficiency, optical modulation, P-N junctions, silicon modulators.

I. INTRODUCTION

BENEFITING from the capabilities of low cost and high integration density, silicon-based photonic integrated chips play an important role in the realization of high-speed and large capacity optical communication systems [1], [2]. Silicon optical modulator [3]–[5] is one of the key devices in photonic integrated chips and determines the performance of transmitter. Due to the continuously increasing demand on communication capacity and the limitation of existing heat dissipation technologies, modulators with high speed and high efficiency have attracted wide attentions. Compared with Mach-Zehnder modulator [6]–[8], the micro-ring modulator [9]–[12] has the inherent advantages of compact footprint [13], [14] and low power consumption [15], making it suitable for large scale photonic integration. However, the conventional depleting PN junction [16] widely used in current silicon micro-ring modulators is of low modulation efficiency ($\sim 1.5 \text{ V}\cdot\text{cm}$), limiting further reduction of power consumption.

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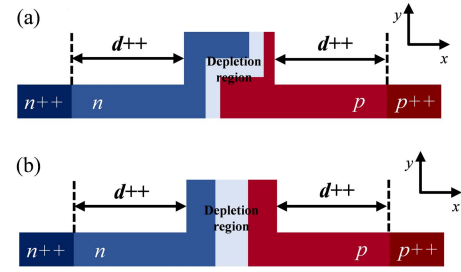


Fig. 1. The waveguide cross-section view of (a) the proposed lateral-zigzag PN junction and (b) the conventional lateral PN junction.

In recent years, several PN junctions with high modulation efficiency have been reported, though they are either insufficient in RC bandwidth or difficult to fabricate. Interleaved [10] and U-shaped [3] silicon PN junctions for ring modulators have been demonstrated with an improved modulation efficiency, while the RC bandwidth is less than 25 Gb/s due to the high capacitance of the junction induced by the large depletion area. A self-aligned wrap around PN junction [17] has been reported for high modulation efficiency and large bandwidth, at the cost of relatively complex fabrication. The fabrication process used to form the doped areas relies on five self-aligned photolithographic steps. L-shaped PN junction consisting of a vertical part and a horizontal part is reported [18], achieving 128 Gb/s PAM4 modulation. More recently, ultra-high speed silicon micro-ring modulators of data rates beyond 200 Gb/s were reported [19], [20]. Nevertheless, there exists a tradeoff between modulation efficiency and bandwidth since the large depletion area for high modulation efficiency will increase the capacitance of the junction, and also the fabrication difficulty.

Here, we propose a novel lateral zigzag PN junction. By enlarging the carrier-light interaction area appropriately, the modulation efficiency is improved and thus a low power consumption is ensured. Simulations show that the modulation efficiency can be doubled while the power consumption can be reduced. Experimental results show that the 3 dB electro-optic bandwidth is 42 GHz and the RC bandwidth is 174 GHz. The modulation power consumption is only 14 fJ/bit for 100 Gb/s PAM4 operation.

II. DEVICE DESIGN AND FABRICATION

Figure 1(a) shows the cross-section view of the proposed lateral-zigzag PN junction. The junction is inserted into a

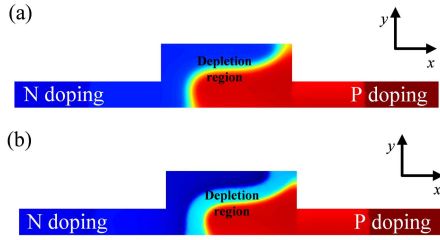


Fig. 2. The carrier concentration profiles of the proposed lateral-zigzag PN junction at a bias voltage of (a) 0 V and (b) -2 V.

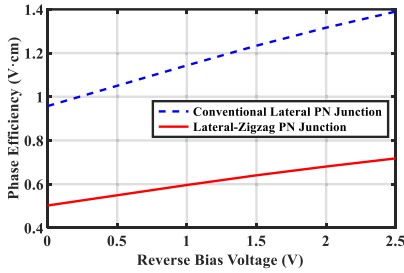


Fig. 3. Simulated phase efficiency ($V_\pi \cdot L_\pi$) under different reverse bias voltage of conventional lateral PN junction and lateral-zigzag PN junction at the wavelength of 1550 nm.

220 nm-high by 450 nm-wide ridge waveguide with a 90 nm-thick slab. The depletion region is composed of a vertical PN junction in the middle and two horizontal PN junctions on both sides. Because the ridge waveguide is asymmetric in the y direction, the energy distribution of the optical field along y direction is non-uniform. The vertical PN junction is specially designed to be located at the position where the energy of the optical field is maximum to achieve an enhanced carrier-light interaction. Furthermore, the P-type region is chosen to locate at the center of the waveguide and covered by the N-type region to obtain an optimal carrier-light spatial overlap, as the carrier dispersion effect of the hole is stronger than that of the electron [5].

Figures 2(a) and 2(b) show the carrier concentration profiles of the proposed lateral-zigzag PN junction at a bias voltage of 0 V and -2 V. The blue regions are N-type and red regions are P-type. The large change of the depletion region area in the middle of the ridge waveguide with increasing reverse bias voltage indicates a high modulation efficiency and large overlap between depletion region and optical mode.

For comparison, the conventional lateral PN junction (Fig. 1(b)) is simulated and analyzed. The distance d between the heavily doped region and the rib is fixed as 400 nm for both junctions. The relationship between the phase efficiency ($V_\pi \cdot L_\pi$) of the PN junction and reverse bias voltages V is simulated and presented in Fig. 3. It can be seen that, under a same voltage, the phase efficiency of the proposed PN junction is about half of the conventional lateral PN junction, showing a doubled phase shift while applying into a modulator.

For demonstration, we fabricated a micro-ring modulator utilizing the proposed PN junction on a silicon-on-insulator (SOI) wafer with 2 μm buried oxide and 220 nm top silicon,

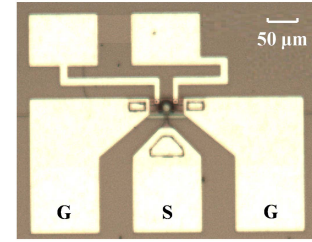


Fig. 4. Optical micrograph of the fabricated micro-ring modulator.

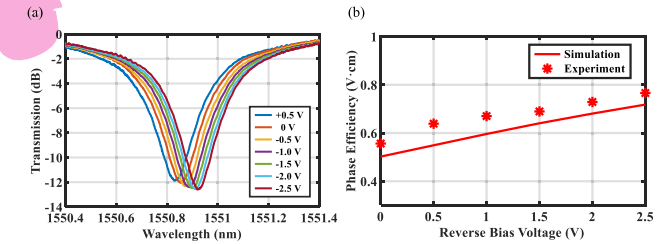


Fig. 5. (a) Measured transmission spectra of the micro-ring modulator with bias voltages varying from +0.5 V to -2.5 V. (b) Measured (asterisks) and simulated (solid line) phase efficiency of the PN junction in the micro-ring modulator.

through deep UV lithography. The micro-ring has a radius of 9 μm and gap size of 0.27 μm , with a quality factor Q of about 7000. The junction is realized by ion implantation, and its length accounts for 70% of the ring circumference. For the proposed lateral-zigzag junction, only one mask is required for P-type and N-type doping in the ridge waveguide region, respectively. Totally, four implantation masks are required for the lateral-zigzag junction, which is the same for a simple lateral PN junction [16].

III. EXPERIMENTS AND DISCUSSIONS

A. DC Characteristics

Figure 4 provides the optical microscope view of the fabricated modulator. The two electrodes above the device are for the on-chip heater to control the operating wavelength of the micro-ring. The on-chip heater is fabricated by depositing titanium nitride (TiN) on the top of the micro-ring modulator [21]. The three lower electrodes in the order of ground-signal-ground are used to load electrical signals onto optical carrier.

Figure 5(a) shows the measured transmission spectra in C band with the bias voltages varying from +0.5 to -2.5 V. The modulation efficiency is 38 pm/V by calculating the variation of resonant wavelength at different voltages. Considering that the zigzag PN junction only occupies 70% of the circumference of the micro-ring, the modulation efficiency can be improved to about 54 pm/V if the entire micro-ring resonator is doped. The modulation efficiency in O band is also measured and calculated to be 23 pm/V. The high modulation efficiency is attributed to the optimized implantation parameters and the novel lateral zigzag PN junction which improves the spatial overlap of depletion area and the optical mode. The measured values of phase efficiency are marked with asterisks and shown in Fig. 5(b), being consistent with the simulated results (the

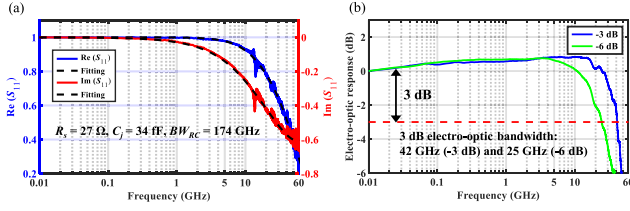


Fig. 6. (a) Measured (solid line) S_{11} parameter of the modulator at a bias voltage of -2.5 V and the fitting curves (dotted line), from which the junction capacitance (34 fF) and resistance (27 Ω) can be extracted. (b) Measured electro-optic response (S_{21}) of the modulator at a bias voltage of -2.5 V with modulation insertion losses of -3 and -6 dB, indicating a 3 dB bandwidth of 42 GHz (-3 dB) and 25 GHz (-6 dB).

solid line). The phase efficiency of the modulator is 0.56 V·cm at a bias voltage of 0 V and 0.76 V·cm at a bias of -2.5 V.

B. Frequency Response

Then, we investigate the frequency response of the designed modulator. The RC bandwidth and electro-optic bandwidth are measured using a vector network analyzer (VNA). The VNA was calibrated to exclude the influence of RF cables and the probe. The insertion loss of the micro-ring modulator is -6 dB, and the applied bias voltage is -2.5 V. We use the model of RC series resistance equivalent circuit to fit the S_{11} response obtained from VNA measurements [16]. The fitted frequency response curve is shown in Fig. 6(a), from which the resistance R_s and junction capacitance C_j of the micro-ring modulator can be extracted as 27 Ω and 34 fF, respectively. The estimated RC-limited electrical bandwidth [9, 16] of the modulator is $BW_{RC} = 174$ GHz.

Figure 6(b) illustrates the measured electro-optic response of the modulator at a bias voltage of -2.5 V with modulation insertion losses of -3 and -6 dB, and the results show that the 3 dB electro-optic bandwidth of the modulator is 42 GHz (-3 dB insertion loss). For the demonstrated micro-ring modulator, the electro-optic bandwidth is mainly limited by the optical bandwidth rather than the RC bandwidth, since the RC bandwidth is much larger than the electro-optic bandwidth. Thus, the electro-optic bandwidth can be further improved by engineering the optical structure of the micro-ring.

C. Modulation of High-Speed Digital Signal

We also carry out the eye-diagram measurements using NRZ and PAM4 data. The experimental setup is shown in Fig. 7. The output of the laser passes through the polarization controller (PC) before being coupled into the on-chip device under test (DUT) via the vertical grating coupler with fundamental transverse electric (TE_0) mode. After modulation, the output light is split by a 10:90 power splitter, with 10% signal being collected by the optical power meter to monitor the modulation insertion loss IL_m , and 90% part being amplified by an erbium-doped fiber amplifier (Amonics AEDFA-33-R-FC) and sampled by an oscilloscope (Keysight N1092A) to observe the eye diagrams. The driving signal of the micro-ring modulator is generated by an arbitrary waveform generator (AWG, Keysight M8196A), followed by an RF amplifier (SHF 807C) and a bias tee (SHF BT65R). The driving voltage swing V_{pp} of

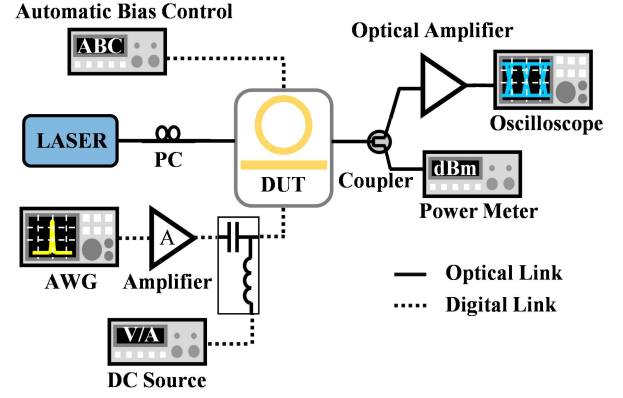


Fig. 7. The measurement setup for high-speed digital signal modulation.

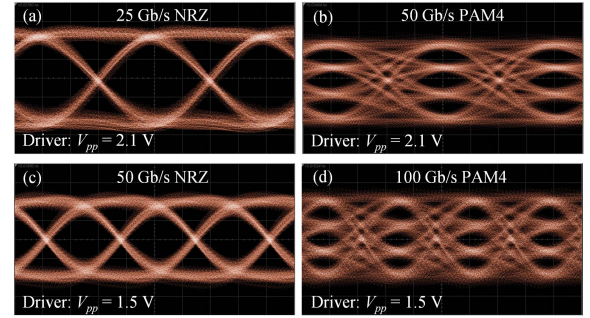


Fig. 8. Measured eye diagrams of (a) 25 Gb/s NRZ at a driving voltage swing of $V_{pp} = 2.1$ V, ER = 5.9 dB, (b) 50 Gb/s PAM4 at $V_{pp} = 2.1$ V, outer ER = 2.3 dB, TDECQ = 1.6 dB, (c) 50 Gb/s NRZ at $V_{pp} = 1.5$ V, ER = 3.9 dB and (d) 100 Gb/s PAM4 at $V_{pp} = 1.5$ V, outer ER = 1.7 dB, TDECQ = 2.9 dB in the C band.

the micro-ring modulator is about 2 V and the bias voltage is -2.5 V, where V_{pp} is the peak-to-peak driving voltage (without any termination resistor). In particular, we use a specially designed automatic bias control circuit (see Fig. 6) to keep the micro-ring operating at the wavelength where the modulation insertion loss IL_m is -6 dB. All the parameters remain the same when measuring the eye diagrams of high speed NRZ and PAM4 signals. In addition, we use pre-emphasis to de-embed the frequency response of RF link before the measurements.

Figures 8(a) and 8(c) show the measured eye diagrams of NRZ signals at 25 and 50 Gb/s in C band, with the driving voltage swing V_{pp} to be 2.1 and 1.5 V, respectively. Similarly, we perform the eye diagrams measurement of 50 and 100 Gb/s PAM4 signals, as shown in Figs. 8(b) and 8(d), indicating that the modulator can achieve single-wavelength modulation up to 100 Gb/s. The energy consumption is calculated to be 14 fJ/bit at a driving voltage swing $V_{pp} = 2.5$ V for the 100 Gb/s PAM4 signals by the following equation [15]:

$$P_{av} = \frac{C_j V_{pp}^2}{N^2 \log_2 N} \sum_{i=1}^{N-1} (N-i) \left(\frac{i}{N-1} \right)^2 \quad (1)$$

where P_{av} is the energy consumed per bit, V_{pp} is the peak-to-peak driving voltage and C_j is the junction capacitance. N is defined as the PAM order. When $N = 2$, the formula calculates the modulation power consumption of the NRZ

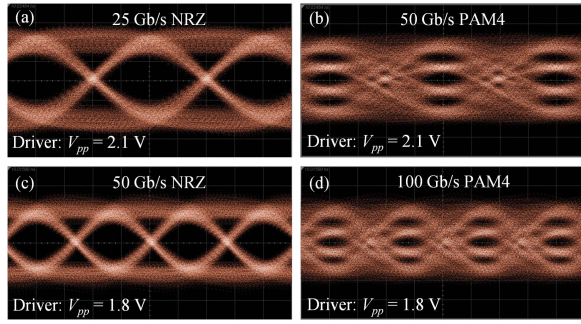


Fig. 9. Measured eye diagrams of (a) 25 Gb/s NRZ at a driving voltage swing of $V_{pp} = 2.1$ V, (b) 50 Gb/s PAM4 at a driving voltage swing of $V_{pp} = 2.1$ V, (c) 50 Gb/s NRZ at a driving voltage swing of $V_{pp} = 1.8$ V and (d) 100 Gb/s PAM4 at a driving voltage swing of $V_{pp} = 1.8$ V in the O band.

TABLE I
PERFORMANCE OF SILICON MICRO-RING
MODULATORS IN RECENT YEARS

Types of PN junctions	Modulation speed (Gb/s)	Phase efficiency (bias voltage) (V-cm)	Modulation efficiency (pm/V)	Power consumption (fJ/bit)	3dB EO Bandwidth (GHz)	Operating wavelength (nm)
U-shape [3]	24 (NRZ)	0.46 (-0.5 V)	40	-	13	1310
Vertical [22]	44 (NRZ)	-	250	17.4	21	1550
Interleaved [11]	44 (NRZ)	1.7	16	300	25	1550
Vertical [14]	100 (PAM4)	0.37	55	-	42	1310
L-shape [18]	128 (PAM4)	0.4 (0 V)	-	18	50	1310
Vertical [20]	200 (PAM4)	0.8	26.34	-	>67	1310
L-shape [19]	192 (PAM4)	0.42 (0 V)	-	5.3	77	1310
L-shape [23]	240 (PAM4)	0.42 (0 V)	-	-	62	1310
Lateral-zigzag (This work)	100 (PAM4)	0.56 (0 V)	38	14	42	1550
		0.92 (0 V)	23			1310

signal. When $N = 4$, the modulation power consumption of PAM4 is calculated. The V_{pp} need to be chosen properly for a tradeoff between power consumption and extinction ratio. The modulation power consumption is proportional to the square of V_{pp} and can be significantly reduced with low V_{pp} .

Figures 9(a)-(d) show the eye diagrams of the micro-ring modulator operating in O band. It can be seen that the eye diagrams quality of C band is better than that of O band, mainly because the optical amplifier in C band has lower noise.

Table I compares the proposed micro-ring modulator with literatures. Due to the large RC bandwidth and high modulation efficiency of the lateral-zigzag junction, our work exhibits low power consumption and superior modulation speed in both O and C bands among these modulators.

IV. CONCLUSION

In summary, we have proposed a high-efficiency silicon micro-ring modulator by applying a novel lateral-zigzag PN junction with enlarged electro-optical interaction area and reduced power consumption. Simulations and measurements show that the junction has superior metrics. A clear and open eye diagram of 100 Gb/s PAM4 signal are successfully achieved with a low modulation power consumption of

only 14 fJ/bit. This work shows the potential of this lateral-zigzag PN junction for low-power and high-speed on-chip modulators.

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